NAME: NIKHIL K R SRN :PES2UG25CS815

SECTION : G

DISASTER RELIEF SUPPLY & VOLUNTEER COORDINATOR

Use-Case Flow: Match and Allocate Supplies

Use Case ID
UC-02

Use Case Name
Match and Allocate Supplies

Primary Actor
Disaster Manager

Supporting System
Disaster Relief Supply & Volunteer Coordinator Platform

Goal
To match an incoming shelter supply requisition against available central warehouse stock, assign it a delivery priority, and route it to the logistics manifest.

Preconditions

1. The disaster relief platform is available.
2. A shelter supply requisition has already been submitted (via UC-01) and is pending in the system.
3. The Disaster Manager is logged in and has access to the central warehouse inventory feed.

Main Success Scenario

1. The Disaster Manager opens the list of pending shelter supply requisitions.
2. The Disaster Manager selects a requisition to process.
3. The system retrieves the requisition details (item category, quantity, shelter location, urgency).
4. The system checks current warehouse stock levels for the requested item(s).
5. The system matches the requisition against available warehouse supply and computes a delivery priority based on need severity, with water and medical items ranked above non-critical items.
6. The system allocates the matched supply to the requisition and routes it to the logistics manifest.
7. The system displays a confirmation to the Disaster Manager showing the assigned priority and manifest entry.
8. The use case ends successfully.

Alternate Flow

A1. Insufficient Warehouse Stock

1. At Step 4, the system detects that available warehouse supply is less than the requisitioned quantity.
2. The system flags the requisition as Partially Fulfillable or Unfulfillable.
3. The system displays the shortfall to the Disaster Manager.
4. The Disaster Manager approves a partial allocation, or escalates the requisition for external resupply.

5. If a partial allocation is approved, the use case resumes at Step 6 with the adjusted quantity.
6. If escalated, the use case ends unsuccessfully.

A2. Warehouse System Unreachable

1. At Step 4, the system cannot connect to the Warehouse System to verify stock.
2. The system retries the connection using cached inventory data if available.
3. If cached data is unavailable or stale beyond the allowed threshold, the system rejects the match attempt.
4. The system notifies the Disaster Manager of the connectivity failure.
5. The use case ends unsuccessfully.

Postconditions

Success
The requisition is allocated warehouse supply, assigned a delivery priority, and routed to the logistics manifest.

Failure
The requisition remains unmatched or is flagged as unfulfillable, and no supply is deducted from warehouse inventory.